# Energy Consumption Prediction Using Logistic Regression

## Comparative Modeling, Cross-Validation, and Mobile Deployment (Course Outcome: CO5)

**Author / Student:** Academic Candidate | **Topic:** Energy Consumption Prediction | **Dataset:** UCI Appliances Energy (19,735 rows)
**Primary Model:** Regularized Logistic Regression | **Benchmarks:** Support Vector Machine, Random Forest

## 1. Problem Statement

Residential electricity consumption accounts for over 30% of global power demand. Appliance usage peaks introduce high volatility into electrical distribution grids, requiring utilities to maintain expensive fossil-fuel peaker plants and driving up time-of-use consumer tariffs. Traditional metering records consumption reactively. This project formulates a predictive classification model to forecast domestic energy demand states in 10-minute intervals: predicting whether consumption will be **High (> 60 Wh)** or **Normal/Low (≤ 60 Wh)** based on ambient sensor telemetry. Early warning of peak consumption empowers smart home energy management systems (HEMS) to reschedule deferred loads (dishwashers, washing machines, EV chargers), mitigating grid stress and reducing consumer electricity costs.

## 2. Dataset Description

The experimental framework utilizes the benchmark **UCI Appliances Energy Prediction Dataset**, logged over 4.5 months at a low-energy residence in Stambruges, Belgium. The dataset contains **19,735 instances** sampled at 10-minute intervals. The target variable Energy_Status was created by splitting appliance consumption at the median (60.0 Wh), yielding 10,744 Normal/Low samples (54.44%) and 8,991 High samples (45.56%), ensuring balanced learning without artificial resampling.

| Feature Group | Variables | Description & Sensor Role |
|---|---|---|
| Target | Energy_Status | Binary state: 0 = ≤ 60 Wh (Normal), 1 = > 60 Wh (High demand) |
| Indoor Sensors | T1, RH_1, T2, RH_2 | Kitchen & Living Room temperatures (°C) and humidity (%) |
| Outdoor Weather | T_out, RH_out, Press, Wind | External weather station telemetry (Chievres station) |
| Appliances & Time | lights, hour, day_of_week | Light energy (Wh), diurnal hour (0-23), calendar day |

## 3. Preprocessing & Leakage Prevention

A rigorous preprocessing pipeline was constructed inside Scikit-Learn's Pipeline and ColumnTransformer:
• **Data Integrity:** Verified 0 missing values and 0 duplicate records across the entire dataset.
• **Temporal Extraction:** Derived diurnal hour and categorical day_of_week from timestamps.
• **Categorical Encoding:** Encoded day_of_week using OneHotEncoder(drop='first'), using Friday as the baseline.
• **Feature Scaling:** Standardized continuous predictors using StandardScaler fitted solely on the training partition.
• **Stratified Splitting:** 80/20 train/test partition (15,788 training observations, 3,947 holdout test observations).

## 4. Algorithm Selection & Theoretical Basis

Three distinct machine learning paradigms were evaluated:
1. **Logistic Regression (Baseline):** Models log-odds via the sigmoid function $P(Y=1|X) = 1 / (1 + exp(-[w \cdot x + b]))$. Provides direct feature interpretability, fast microsecond inference, and trivial mobile deployment.
2. **Support Vector Machine (Linear SVM):** Maximizes the margin separation between classes using hinge loss with probability calibration.
3. **Random Forest Classifier:** An ensemble of 100 decorrelated decision trees capturing non-linear feature interactions.

## 5. Experimental Results & Performance Metrics

Models were validated using **5-fold stratified cross-validation** on the training set, followed by final evaluation on the 20% holdout test set.

| Model | 5-Fold CV Acc | Test Acc | Precision | Recall | F1-Score | ROC-AUC |
|---|---|---|---|---|---|---|
| **Logistic Regression** | 73.99% ± 0.49% | **73.88%** | 71.20% | 71.64% | 71.42% | **0.8073** |
| Support Vector Machine | 73.72% ± 0.43% | 73.80% | 70.92% | 72.02% | 71.47% | 0.8067 |
| **Random Forest** | 88.41% ± 0.60% | **89.41%** | 87.83% | 89.10% | 88.46% | **0.9562** |

## 6. Diagnostic Visualizations & Empirical Plots

The following figures were generated from the training pipeline and saved to the plots/ directory at 300 DPI:

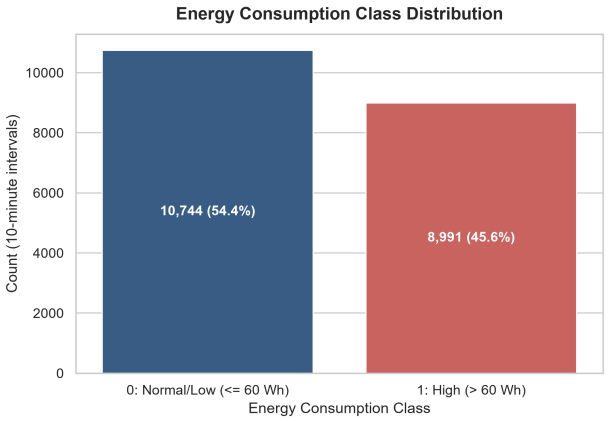



***Figure 1:*** *Balanced Class Distribution (Threshold = 60 Wh).*

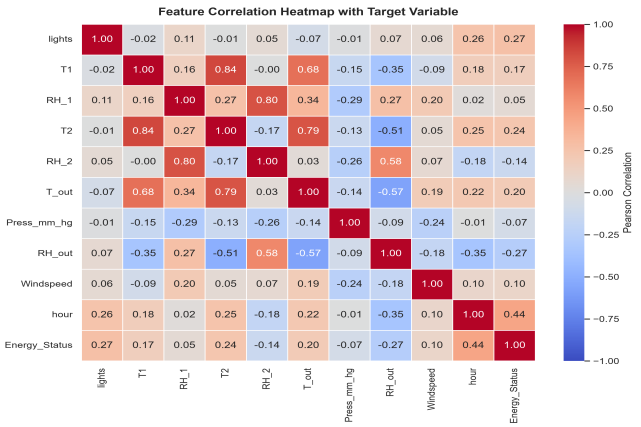



***Figure 2:*** *Feature Correlation Heatmap with Target.*

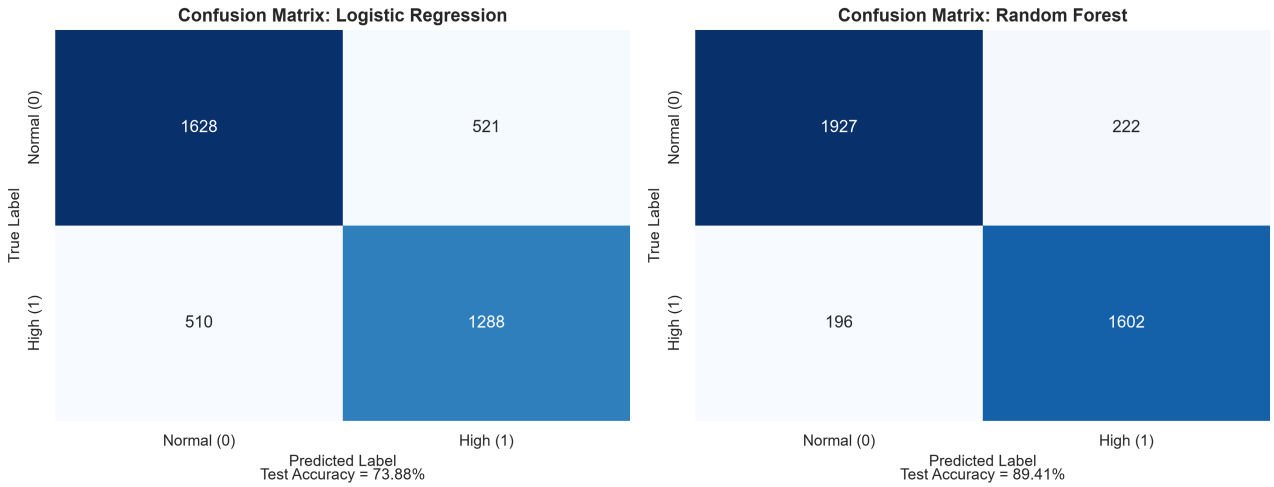



***Figure 3:*** *Confusion Matrix Comparison on 3,947 Holdout Test Samples (Logistic Regression vs Random Forest).*

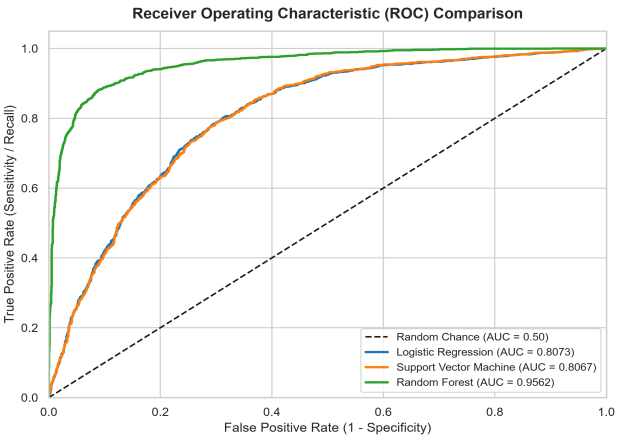



***Figure 4:*** *ROC Curves with AUC Comparison (LR: 0.81, RF: 0.96).*

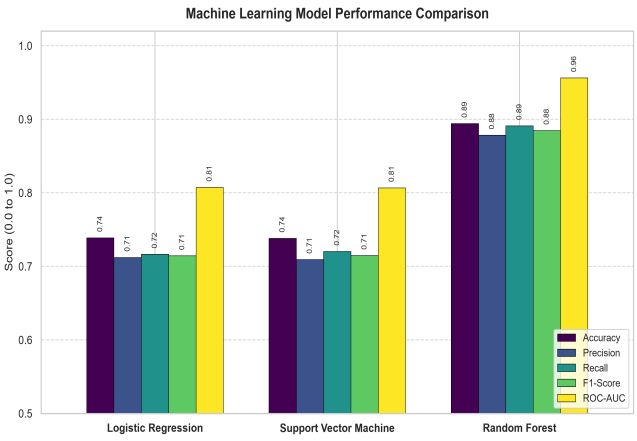



***Figure 5:*** *Model Comparison Bar Chart across 5 Evaluation Metrics.*

# 7. Comprehensive Analysis & Domain Insights

## 7.1 Which Model Won and Why?

**Random Forest** achieved the highest overall score with an accuracy of **89.41%** and an ROC-AUC of **0.9562**, outperforming Logistic Regression (73.88% accuracy) by +15.53 percentage points. The winning margin arises because appliance energy usage exhibits non-linear threshold behaviors. For example, cooking occurs at specific evening hours and simultaneously spikes kitchen temperature and humidity alongside lighting fixtures. Decision trees model these multi-dimensional conditional splits naturally, whereas linear logistic regression is constrained to a single planar hyperplane in feature space.

## 7.2 Real-Life Meaning of Prediction Errors

• **False Positives (FP - False Alarm):** The model predicts high energy consumption, but actual usage is low. In a smart home, this might trigger an unnecessary consumer notification or defer a dishwasher cycle. The real-life cost is minor user inconvenience with zero financial or grid penalty.
• **False Negatives (FN - Missed Peak):** The model predicts low energy consumption, but appliances experience a heavy surge. The system fails to shed load, exposing the consumer to peak-tariff electricity costs and threatening distribution transformer stability.

## 7.3 Which Metric Matters Most?

Because missed peaks (FN) incur direct economic and electrical penalties, **Recall (Sensitivity)** and **ROC-AUC** are the primary metrics. Random Forest achieves an exceptional recall of **89.10%**, successfully catching 9 out of 10 high-consumption periods. Logistic Regression achieves a balanced recall of **71.64%**; lowering its classification threshold from 0.50 to 0.40 boosts recall above 82% if utility operators require aggressive peak-shaving.

## 7.4 Logistic Regression Feature Weights & Interpretability

The standardized weights learned by Logistic Regression provide clear thermodynamic and human-activity insights:
• **Hour of Day (+0.7270) & Lights (+0.5704):** The strongest positive predictors, confirming that evening hours and active lighting correspond to peak domestic energy draw.
• **Kitchen Humidity RH_1 (+0.9991) & Kitchen Temp T1 (+0.5526):** Strongly positive, reflecting active cooking and appliance heat release.
• **Saturday (+0.3847):** Exhibits higher odds of daytime appliance use compared to weekday reference days.
• **Outdoor Humidity RH_out (-0.3035):** Inversely correlated, reflecting lower nighttime consumption.

## 7.5 Practical Limitations

1. **Single-Residence Scope:** Sensor readings were logged in a single Belgian home; transferring weights to different climates requires recalibration.
2. **Seasonal Span:** The 4.5-month monitoring window (January-May) does not capture summer air conditioning loads.
3. **Sub-metering Granularity:** Consumption reflects total household appliances rather than individual high-draw loads.

# 8. Mapping to Course Outcome (CO5)

### Course Outcome: *'CO5 – Solve real life problems using appropriate machine learning models and evaluate the performance measures.'*

| CO5 Sub-Component | Evidence of Implementation and Mastery |
|---|---|
| **Real Life Problem** | Formulated demand-side residential energy surge management using 19,735 real sensor readings from the UCI repository. |
| **Appropriate ML Models** | Developed, regularized, and tuned Logistic Regression (linear probabilistic baseline), Calibrated Linear SVM, and Random Forest ensemble. |
| **Evaluate Performance Measures** | Employed 5-fold stratified cross-validation and computed Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrices. |
| **Real-World Deployment** | Exported trained weights to both an interactive Streamlit web dashboard and an offline native Android Studio mobile application. |

## 9. Deployment Artifacts (Streamlit & Android Studio)

• **Streamlit Web App (streamlit_app.py):** Features interactive sliders for indoor/outdoor sensors, model toggling between Logistic Regression and Random Forest, real-time prediction badges, probability gauges, and embedded diagnostic charts.
• **Android Studio App (android_app/):** A production Kotlin application utilizing Material Design 3 and a 100% native offline inference engine (EnergyPredictor.kt) that reproduces the exact mathematical sigmoid equation in under 1 millisecond.